

LD15 range high intensity LED beacon

Ex db, weatherproof



Overview

These certified LED beacons have been designed for use in potentially explosive atmospheres and harsh environmental conditions.

The enclosures are suitable for use offshore or onshore where light weight combined with corrosion resistance is required.

The housings are manufactured completely from a UV stable glass reinforced polyester. Stainless steel screws and mounting bracket are incorporated ensuring a totally corrosion free product.

The LD15 incorporates a low maintenance, innovative LED array, which has a life of up to 54,000 hours.

The low current consumption design allows an energy efficient solution without compromising light output. Units can be painted to customers specification and supplied with identification labels.

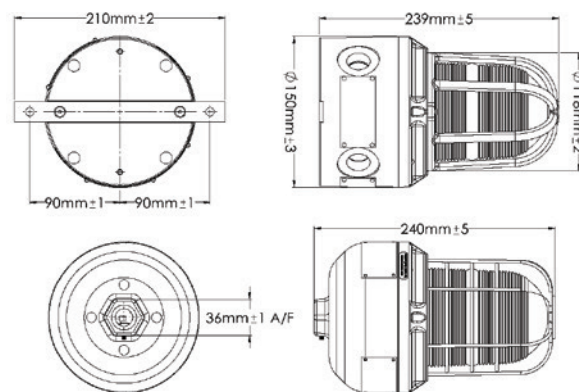
Features

- Zone 1, 2, 21 & 22
- Ex db IIC T6/T5/T4
- ATEX Certified Ex II 2GD
- IECEx Certified Gb, Db
- UL certified for USA and Canada
 - Hazardous locations:
 - Class I, Div. 2, Groups A, B, C & D
 - Class II, Div. 2 Groups F & G
 - Class I, Zone 1, AExd, IIC T6/T5
 - Ordinary locations: visual-signal device
- CCC Ex certified
- INMETRO certified
- IP66 & 67
- SIL 2 certified
- Certified temperature -55°C to +85°C
- Corrosion-free GRP
- Optional stainless steel backstrap
- Various lens colours
- Optional cast aluminium or stainless steel wire guard
- Up to 3 x M20 or 3 x M25 entries
- 5 user selectable functions:
 - 60 fpm
 - 80 fpm
 - 120 fpm
 - Double flash
 - Steady
- Remote and local function selection



Certifications	
ATEX Ex db	Cert. no. Baseefa 04ATEX0009X Certified to: EN IEC 60079-0, EN 60079-1, EN 60079-31 Ex II 2GD Ex db IIC T4/T5/T6 Gb, Ex tb IIIC T135°C/T100°C/T85°C Db
IECEX Ex db	Cert. no. IECEX BAS 05.0048X Certified to: IEC60079-0, IEC60079-1, IEC60079-31 Ex db IIC T4/T5/T6 Gb, Ex tb IIIC T135°C/T100°C/T85°C Db
INMETRO Ex db	Ex db IIC T4/T5/T6 Gb, Ex tb IIIC T135°C/T100°C/T85°C Db
CCC Ex	Ex db IIC T5/T6 Gb, Ex tb IIIC T85°C/T100°C Db
GB (GuoBiao)	Ex db op is IIC T5/T6 Gb, Ex tb op is IIIC T85°C/T100°C Db
UL Haz Locs	Cert. no. E187894
UL Ord Locs/Marine	Cert. no. S8128. Visual signal device
ULC	Cert. no. E187894
UL certified	Class I, Div. 2, Groups A, B, C & D Class II, Div. 2, Groups F & G Class I, Zone 1, AEx d IIC T5/T6
SIL	SIL 2 certified to IEC61508. Cert number FSP 13005
Specifications	
Material	Body: glass reinforced polyester Lens: borosilicate glass Backstrap: stainless steel 316 Wire guard (optional): stainless steel wire Cast guard (optional): aluminium LM25M
Finish	Natural black or painted to customer specification
Voltage	12V d.c. to 48V d.c., 110V a.c. to 254V a.c
LED life	Up to 54,000 hours
Function	Steady, 60, 80 & 120 fpm, double flash
Certified temp ATEX, IECEX, INMETRO	-55°C to +85°C (T4) -55°C to +70°C (T5) -55°C to +55°C (T6)
Certified temp UL, CCC Ex + GB, SIL 2	-55°C to +70°C (T5) -55°C to +55°C (T6)
Weight	Pipe mount: 3.1kg Direct mount: 3.5kg
Ingress protection	IP66 & IP67
Entries	ATEX version: Standard: 2 x M20 Optional: 1 x M20/M25, 3 x M20/M25 entries UL version: Standard pipe: ¾" NPT Standard direct: 2 x ¾" NPT Optional direct: 1 x ½"/¾" NPT or 3 x ½"/¾" Dual UL/ATEX, UL/IECEX: Standard direct: 2 x ¾" NPT
Terminals	Direct mount: 12 x 2.5mm²/14AWG Pipe mount: 8 x 14AWG
Labels	Tag/duty optional
EOL Resistor	Customers may fit a monitoring resistor to DC units. The maximum total power dissipation through the resistor must be limited to 2W.

General arrangement drawing (all dimensions in mm)



Electrical:

Nominal voltage	d.c.			a.c. (50Hz*)				
	12	24	48	110	120	230	240	254
Average current steady (mA)	430	220	120	70	70	40	40	40
Average current 60fpm (mA)	880	420	220	90	90	50	50	50
Average current 80fpm (mA)	880	420	220	90	90	50	50	50
Average current 120fpm (mA)	880	420	220	90	90	50	50	50
Average current double (mA)	710	340	180	40	40	30	30	30

*For 60Hz please refer to the US datasheet

Typical light output (effective Cd) (no guard)

Colour	Red	Blue	Green	Amber	Clear	Yellow
60 fpm	61	24	86	55	128	122
80 fpm	55	22	78	49	117	112
120 fpm	47	19	66	42	100	95
Steady*	41	17	59	39	86	82

*Not available ULC

Ordering requirements

The following code is designed to help in selection of the correct unit. Build up the reference number by inserting the code for each component into the appropriate box

Model	Certification	Voltage	Lens colour	Lens guard	Fixings	Options	Finish																																																																																										
LD15																																																																																																	
	<table> <tr> <th>Certification</th><th>Code</th></tr> <tr><td>ATEX</td><td>B</td></tr> <tr><td>IECEX</td><td>J</td></tr> <tr><td>Dual ATEX/UL</td><td>AU</td></tr> <tr><td>Dual IECEX/UL</td><td>JU</td></tr> <tr><td>UL</td><td>UL</td></tr> <tr><td>UL (ordinary location)</td><td>UW</td></tr> <tr><td>ULC</td><td>UC</td></tr> <tr><td>INMETRO</td><td>DM</td></tr> <tr><td>CCC Ex/GB</td><td>Q</td></tr> <tr><td>Weatherproof</td><td>W</td></tr> </table>	Certification	Code	ATEX	B	IECEX	J	Dual ATEX/UL	AU	Dual IECEX/UL	JU	UL	UL	UL (ordinary location)	UW	ULC	UC	INMETRO	DM	CCC Ex/GB	Q	Weatherproof	W	<table> <tr> <th>Voltage</th><th>Code</th></tr> <tr><td>12-48Vdc</td><td>DC</td></tr> <tr><td>110-254Vac</td><td>AC</td></tr> </table>	Voltage	Code	12-48Vdc	DC	110-254Vac	AC	<table> <tr> <th>Lens colour</th><th>Code</th></tr> <tr><td>Red</td><td>R</td></tr> <tr><td>Blue</td><td>B</td></tr> <tr><td>Green</td><td>G</td></tr> <tr><td>Amber</td><td>A</td></tr> <tr><td>Clear</td><td>C</td></tr> <tr><td>Yellow</td><td>Y</td></tr> </table>	Lens colour	Code	Red	R	Blue	B	Green	G	Amber	A	Clear	C	Yellow	Y	<table> <tr> <th>Guard</th><th>Code</th></tr> <tr><td>None</td><td>N</td></tr> <tr><td>Cast</td><td>C</td></tr> <tr><td>Wire</td><td>W</td></tr> </table>	Guard	Code	None	N	Cast	C	Wire	W	<table> <tr> <th>Fixings</th><th>Code</th></tr> <tr><td>Pipe mount</td><td>P*</td></tr> <tr><td>Direct mount without backstrap</td><td>D†</td></tr> <tr><td>Direct mount with backstrap</td><td>B†</td></tr> </table> *Only available on UL listings †Contact MEDC if ½"NPT is required	Fixings	Code	Pipe mount	P*	Direct mount without backstrap	D†	Direct mount with backstrap	B†	<table> <tr> <th>Option</th><th>Code</th></tr> <tr><td>None</td><td>N</td></tr> <tr><td>Tag label</td><td>T*</td></tr> <tr><td>Duty label</td><td>D*</td></tr> <tr><td>Relay initiate</td><td>R†**</td></tr> <tr><td>Telephone initiate</td><td>I‡**</td></tr> <tr><td>Blanking plug</td><td>P</td></tr> <tr><td>EOL Resistor</td><td>E§</td></tr> </table> *Please specify. For UL one label only. For AU/JU no labels. †Suitable for 24Vdc initiate supplies only ‡Not available on UL/UW/ULC **Not available on SIL § Please specify	Option	Code	None	N	Tag label	T*	Duty label	D*	Relay initiate	R†**	Telephone initiate	I‡**	Blanking plug	P	EOL Resistor	E§	<table> <tr> <th>Finish</th><th>Code</th></tr> <tr><td>Natural black</td><td>N</td></tr> <tr><td>Red</td><td>R</td></tr> <tr><td>Blue</td><td>B</td></tr> <tr><td>Yellow</td><td>Y</td></tr> <tr><td>Green</td><td>G</td></tr> <tr><td>White</td><td>W</td></tr> <tr><td>Special</td><td>S*</td></tr> </table> *Please specify	Finish	Code	Natural black	N	Red	R	Blue	B	Yellow	Y	Green	G	White	W	Special	S*
Certification	Code																																																																																																
ATEX	B																																																																																																
IECEX	J																																																																																																
Dual ATEX/UL	AU																																																																																																
Dual IECEX/UL	JU																																																																																																
UL	UL																																																																																																
UL (ordinary location)	UW																																																																																																
ULC	UC																																																																																																
INMETRO	DM																																																																																																
CCC Ex/GB	Q																																																																																																
Weatherproof	W																																																																																																
Voltage	Code																																																																																																
12-48Vdc	DC																																																																																																
110-254Vac	AC																																																																																																
Lens colour	Code																																																																																																
Red	R																																																																																																
Blue	B																																																																																																
Green	G																																																																																																
Amber	A																																																																																																
Clear	C																																																																																																
Yellow	Y																																																																																																
Guard	Code																																																																																																
None	N																																																																																																
Cast	C																																																																																																
Wire	W																																																																																																
Fixings	Code																																																																																																
Pipe mount	P*																																																																																																
Direct mount without backstrap	D†																																																																																																
Direct mount with backstrap	B†																																																																																																
Option	Code																																																																																																
None	N																																																																																																
Tag label	T*																																																																																																
Duty label	D*																																																																																																
Relay initiate	R†**																																																																																																
Telephone initiate	I‡**																																																																																																
Blanking plug	P																																																																																																
EOL Resistor	E§																																																																																																
Finish	Code																																																																																																
Natural black	N																																																																																																
Red	R																																																																																																
Blue	B																																																																																																
Yellow	Y																																																																																																
Green	G																																																																																																
White	W																																																																																																
Special	S*																																																																																																